UCC601: GEOTECHNICAL ENGINEERING-II

L	T	P	Cr
3	0	2	4.0

Course Objective: This subject aims to expose the students to geotechnical design of different types of shallow and deep foundations. Further they will be expose to understanding of earth pressure for stability of retaining structures along with various techniques for stability of slopes.

Soil Exploration: Introduction to soil exploration, scope, soil exploration for different structures, spacing, significant depth, boring and sampling techniques, bore hole plan, types of samples, penetration test (SCP and SPT), sample disturbances and Geophysical methods.
Earth Pressure: At rest condition, states of plastic equilibrium, Rankine and Coulomb's theories for active and passive conditions, Lateral Earth pressure in layered soils. Stability of cantilever Retaining wall.
Stability of Slopes: Infinite slope, types of failure, total and effective stress analysis, Taylor's stability numbers, concept of factors of safety, method of slices, modified method of slices, Swedish's circle method.
Bearing Capacity: Definitions, introduction to shallow and deep foundation, depth of foundation, Concept of net and gross bearing capacity. Terzaghi's general bearing capacity equation, IS code equation, factors affecting bearing capacity. Settlements for clays and sands, permissible settlements, bearing capacity by penetration tests, Influence of eccentric and inclined loads, plate load test.
Pile Foundations: Types, function, selection of piles, pile driving formulae, equipment, point, bearing and friction piles. Load carrying capacity of single pile, group action, spacing of piles, Negative skin friction, settlement of pile groups, under-reamed piles.
Caissons and Wells: Introduction, components, shapes, Construction and sinking of well, tilts and shifts.

Laboratory Work:
The students will be introduced to various laboratory & field experiments as per BIS specifications.

1. Determination of bearing capacity by Standard Penetration test.
2. Determination of bearing capacity by Plate load test.
3. Determination of bearing capacity by static and dynamic cone Penetration test.
4. Numerical modelling and simulation of earth retaining systems
5. Numerical modelling and simulation of slopes,
6. Numerical modelling and simulation of shallow foundation systems,
7. Numerical modelling and simulation of deep foundation systems

Course learning Outcome:
After completion of this course, the students will be able to:

1. Design and analyze problems related to shallow and machine foundations
2. Analyze lateral earth pressure for design of earth retaining structures
3. Assess stability of natural/man-made slopes under varying in-situ material properties
4. Design and analyze problems related to pile and well foundations

Text Books

1. *Gopal Ranjan& A.S.R. Rao, Basic and Applied Soil mechanics, New Age Publisher, New Delhi (2016)*
2. *Murthy V.N.S., Advanced Foundation Engineering, C.B.S. Publisher, New Delhi .(2017)*
3. *Parshotham Raj, Geotechnical Engg., Pearson , New Delhi .(2013)*

References Books

1. *Bowles J.E., Foundation Analysis and Design, McGraw Hill Book Company, New York .(2009)*
2. *Das B.M., Principles of Soil Mechanics, Thomson Publisher, USA. (2015)*
3. *Saran Swami, Soil Dynamics and Machine Foundations, Galgotia Publishers, New Delhi .(2016)*

Evaluation Scheme:

Sr. No.	**Evaluation Elements**	**Weights (%)**
1.	MST	30
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30